

DSAgent

Autonomous Pipeline Report

Dataset: titanic.csv

Generated: May 19, 2026 at 10:45 PM

Session: 05de9ee9-496...

21

Steps Executed

6

Phases Completed

59.8s

Total Time

1. Dataset Overview

The dataset contains **891** rows and **12** columns, using **0.28 MB** of memory.

Type	Count	Columns
Numeric	7	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Categorical	5	Name, Sex, Ticket, Cabin, Embarked

Numeric Statistics

Column	Mean	Median	Std	Min	Max
PassengerId	446.00	446.00	257.35	1.00	891.00
Survived	0.38	0.00	0.49	0.00	1.00
Pclass	2.31	3.00	0.84	1.00	3.00
Age	29.70	28.00	14.53	0.42	80.00
SibSp	0.52	0.00	1.10	0.00	8.00
Parch	0.38	0.00	0.81	0.00	6.00
Fare	32.20	14.45	49.69	0.00	512.33

AI Analysis

Key Insights about the Dataset

- Dataset Overview: The dataset contains 891 rows and 12 columns, with a mix of numeric and categorical data. It's relatively small and lightweight, using only 0.28 MB of memory.
- Numeric columns: PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
- Categorical columns: Name, Sex, Ticket, Cabin, Embarked
- Survival Rate: The survival rate is 38%, indicating that the dataset is likely from the Titanic dataset, where survival was influenced by factors like class, gender, and age.
- Age Distribution: The average age is 29.7 years, with a wide range from 0.42 to 80 years. There are significant missing values in the Age column (19.87%).
- Class Distribution: The majority of passengers were in Pclass 3 (lower class), with a mean of 2.31. There's a moderate negative correlation between Pclass and Survived, suggesting lower class passengers had lower survival rates.
- Fare Distribution: The average fare is \$32.2, with a high variance (standard deviation of \$49.69), and a strong negative correlation with Pclass (suggesting lower class passengers paid less).

Data Quality Issues Found

- High Missing Values:
- Cabin: 77.1% missing values (687 missing) – this is a major issue.
- Age: 19.87% missing values (177 missing) – significant but not as severe as Cabin.
- Embarked: 0.22% missing values (2 missing) – minimal impact.
- High Cardinality in Categorical Columns:
- Name has 891 unique values, making it impractical to use directly in models. It may need to be processed or discarded.

Potential Data Type Issues

- PassengerId is numeric but may be treated as a unique identifier, not a feature.
- Fare has a wide range and is a good candidate for log transformation to reduce skew.

Columns That Look Most Interesting/Useful

- Survived: This is the target variable and is central to the problem (predicting survival).
- Pclass: Strongly correlated with Survived (moderate negative correlation), suggesting it's a key feature. It reflects socioeconomic status.
- Age: Moderate correlation with Survived and Pclass, and is a key demographic factor.
- Sex: Categorical with high frequency (male dominates), and likely has a strong impact on survival.
- Fare: Strong negative correlation with Pclass, and may reflect economic status.
- Embarked: Categorical with low missing values, and may indicate port of embarkation, which could correlate with class or survival.
- SibSp and Parch: These indicate family relationships. They have moderate correlations with other features and may provide insights into

survival behavior (e.g., children or family groups). Likely Target Variable (for ML) • Target Variable: Survived • This is a binary classification task (0 = died, 1 = survived). • It is the central outcome of the dataset and is the most logical target for a predictive model. • The mean of 0.38 indicates a class imbalance (more non-survivors), which should be addressed in modeling (e.g., through resampling or using class weights).

Summary • Key Features to Use: Pclass, Age, Sex, Fare, SibSp, Parch, Embarked • Target Variable: Survived • Data Quality Issues: High missing values in Cabin and Age, high cardinality in Name • Next Steps: • Impute or drop missing values in Age and Cabin. • Encode categorical variables (Sex, Embarked). • Consider dropping Name due to high cardinality. • Explore interactions between Pclass, Age, and Sex for survival prediction.

2. Data Quality Assessment

Found **3** columns with missing values out of 891 rows.

Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

Potential Issues:

- High cardinality categorical columns: ['Name']

3. Data Cleaning Actions

Age has 19.87% missing values

column: Age
strategy: median
action: Filled with median: 28.00
nulls_before: 177
nulls_after: 0
rows_after: 891

Embarked has 0.22% missing values

column: Embarked
strategy: mode
action: Filled with mode: S
nulls_before: 2
nulls_after: 0
rows_after: 891

Fare has high variance and potential outliers

column: Fare
method: iqr
threshold: 1.5
rows_removed: 116
rows_remaining: 775
removal_percentage: 13.02

Cabin has 77.1% missing values and is not useful for analysis

column: Cabin
strategy: drop
action: Dropped 661 rows
nulls_before: 661
nulls_after: 0
rows_after: 114

Age has a wide range and potential outliers

column: Age
method: iqr
threshold: 1.5

```
outlier_count: 1
outlier_percentage: 0.88
total_rows: 114
```

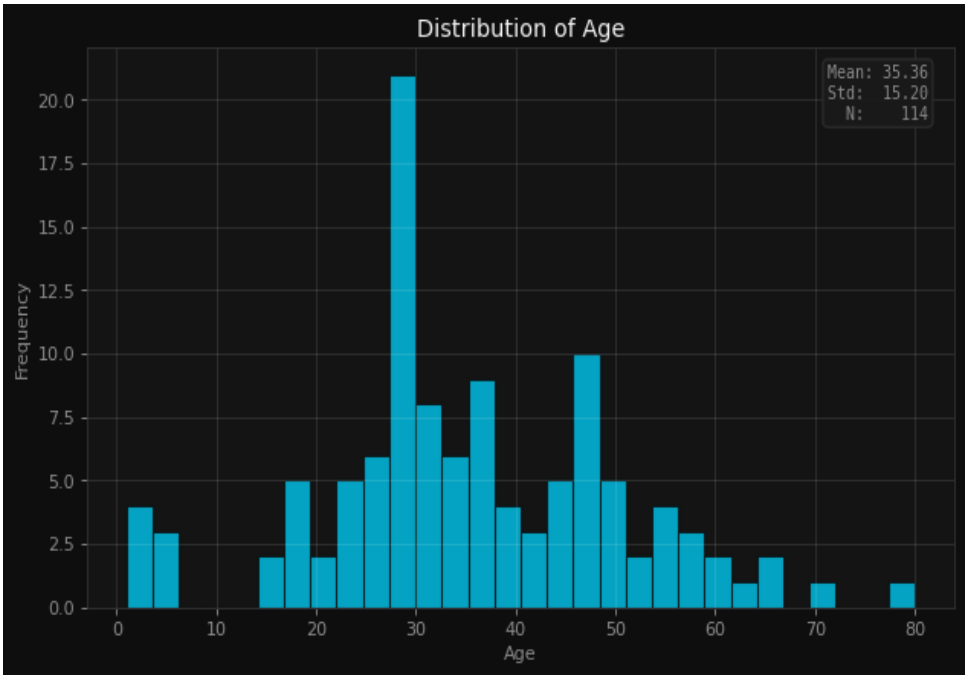
Fare has high variance and potential outliers

```
column: Fare
method: iqr
threshold: 1.5
outlier_count: 0
outlier_percentage: 0.0
total_rows: 114
```

Applied 6 cleaning operations based on data quality analysis.

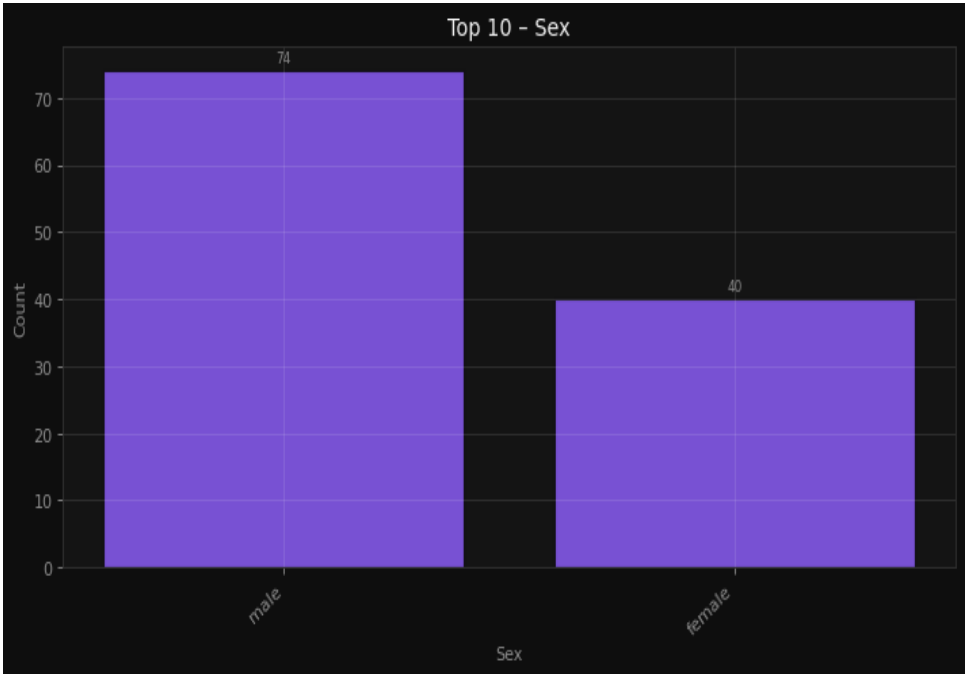
4. Visualizations & Insights

Understand the age distribution of passengers



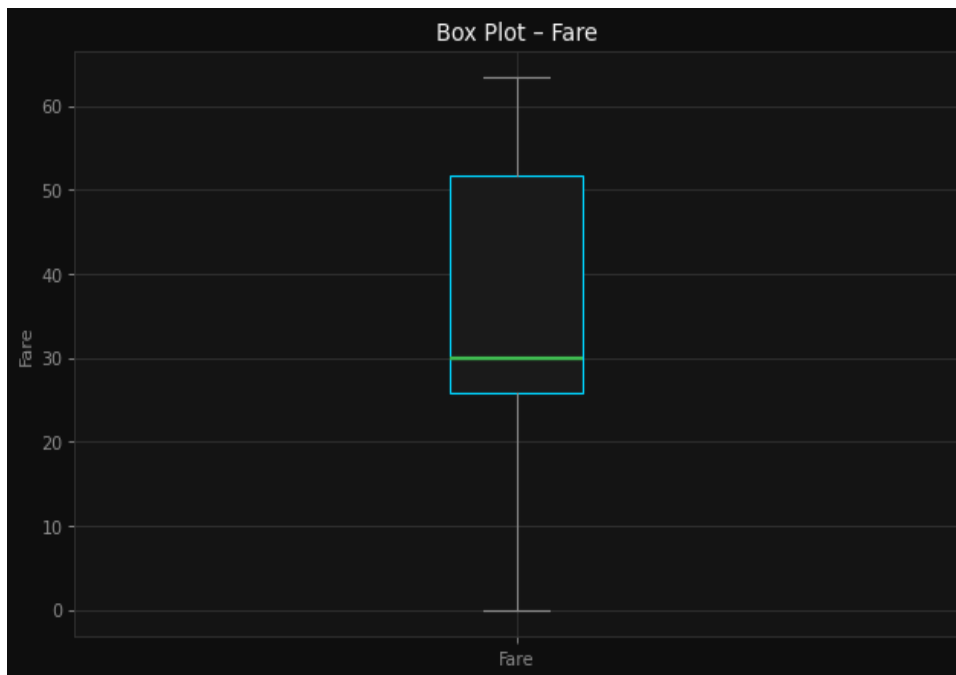
Understand the age distribution of passengers

Analyze the gender distribution



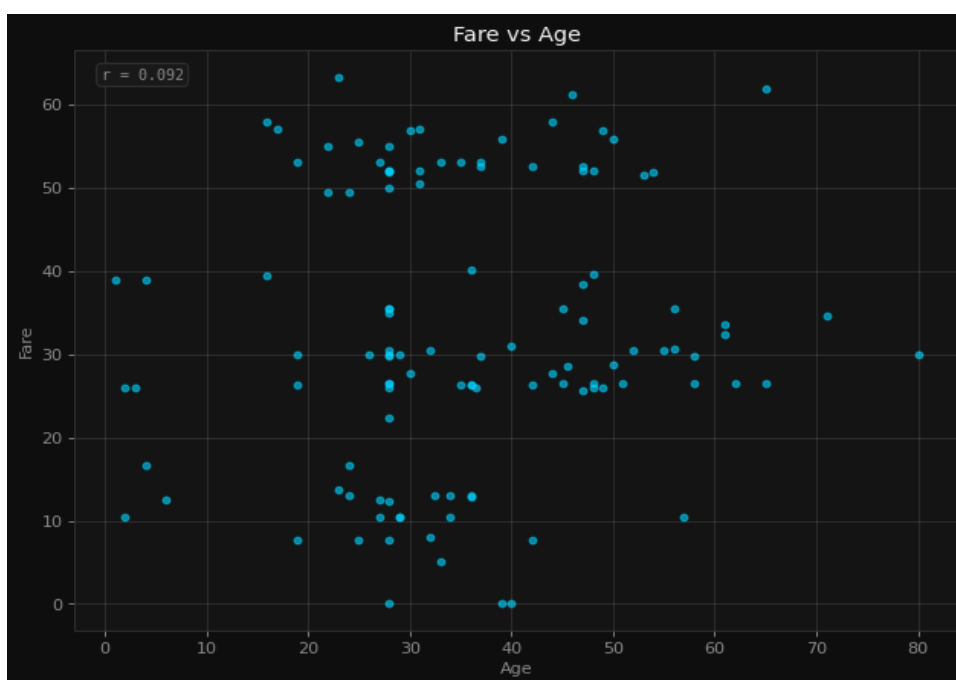
Analyze the gender distribution

Detect outliers in fare payments



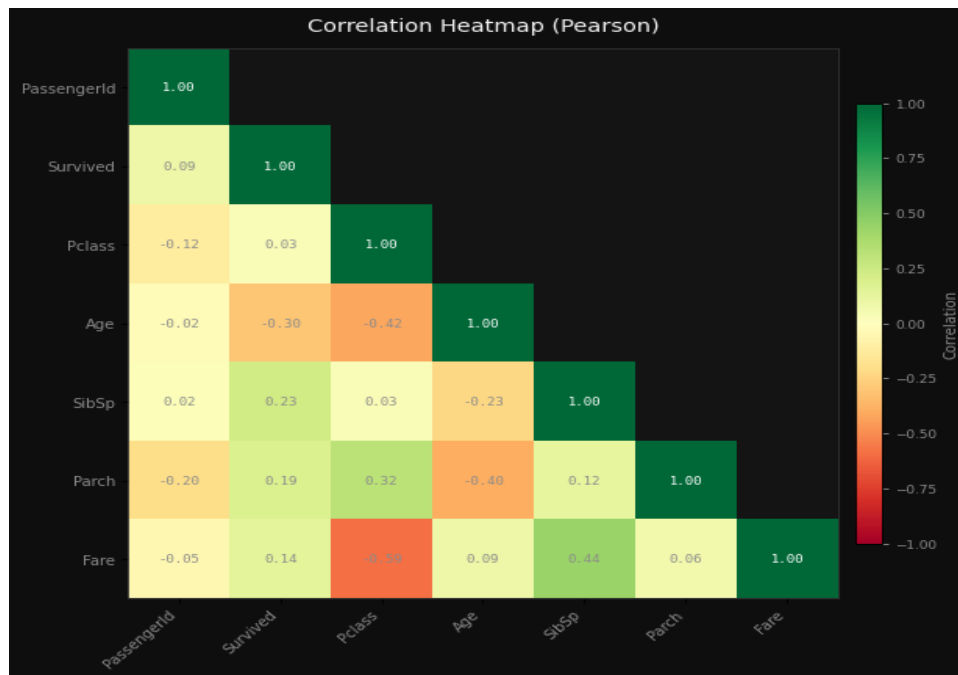
Detect outliers in fare payments

Explore the relationship between age and fare



Explore the relationship between age and fare

Identify correlations between numeric features



Identify correlations between numeric features

AI Interpretation

Histogram (Age Distribution): Likely shows a right-skewed distribution with a peak around 20–30 years, reflecting the typical age range of passengers on the Titanic. The presence of missing values may be visible as gaps or lower frequency in certain age ranges. Bar Chart (Gender Distribution): Likely reveals a significant imbalance, with more male passengers than female, consistent with historical context of the Titanic disaster. Box Plot (Fare Outliers): Likely highlights a few high fare outliers, possibly indicating first-class passengers, while the majority of fares cluster around lower values. Scatter Plot (Age vs. Fare): Likely shows a weak or no clear correlation, with scattered points, suggesting age does not strongly influence fare payment. Correlation Heatmap: Likely reveals low or no correlations between numeric features (e.g., Age and Fare), with Pclass possibly showing a moderate negative correlation with Fare (higher class = higher fare).

5. Feature Engineering

Encode categorical variables for ML modeling

```
encoding: One-Hot (get_dummies)
drop_first: True
columns_before: 12
columns_after: 13
new_columns_added: 1
rows: 114
```

Normalize numeric features for better model performance

```
scaler: StandardScaler (Z-score)
rows: 114
note: Dataset updated in session. Features now have  $\mu \approx 0$ ,  $\sigma \approx 1$ .
```

Reduce skew in Fare distribution

Applied 3 feature engineering steps.

6. Model Training & Comparison

Problem Type: **classification** | Target: **Survived** | Features: **12**

Best Model: **XGBoost** — Score: **69.6%**

Model	Accuracy	Precision	Recall	F1
Random Forest	65.2%	64.2%	65.2%	64.2%
XGBoost	69.6%	69.1%	69.6%	69.2%
LightGBM	56.5%	56.5%	56.5%	56.5%
Logistic Regression	60.9%	61.7%	60.9%	61.2%

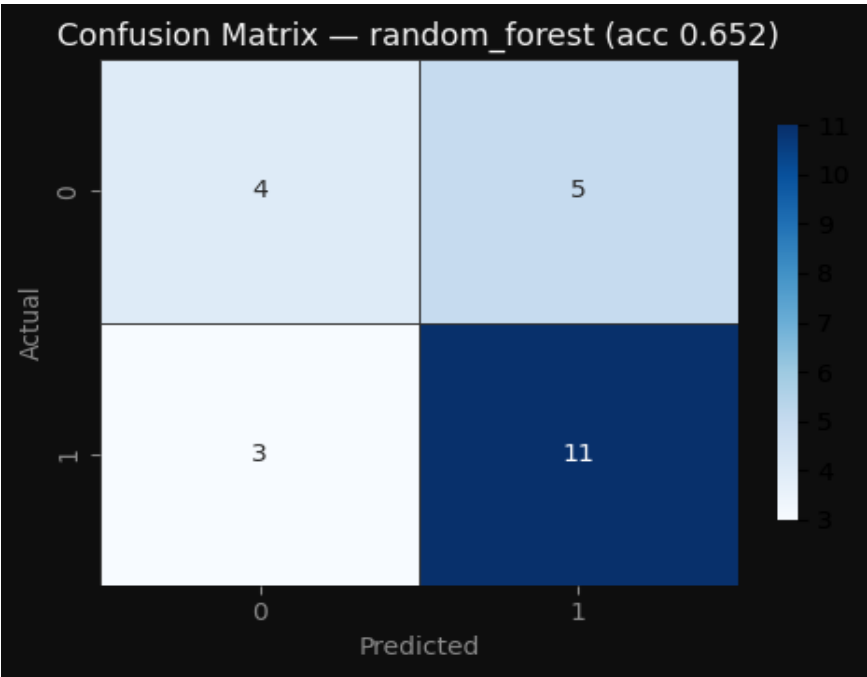
Model Selection Rationale

LLM error: 402

7. Model Evaluation

Model Evaluation Metrics

accuracy: 65.2% | precision: 64.2% | recall: 65.2% | f1_score: 64.2%



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

LLM error: 402